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A Decision Framework for Clean Energy Supply Chain Optimization

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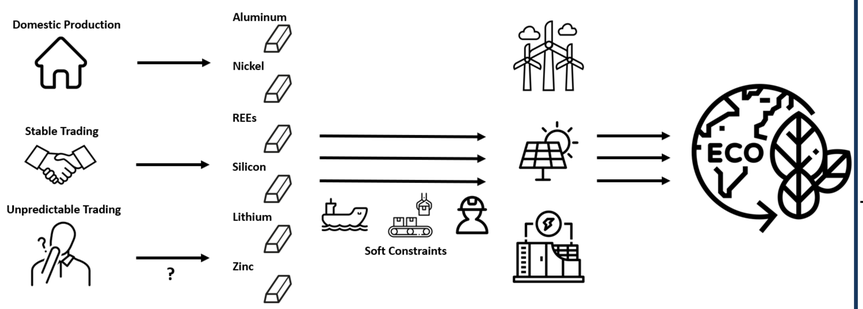
Introduction and Motivation

The clean energy transition is heavily dependent on the effective management of supply chains, particularly the availability of critical materials for manufacturing.



- Rising material demand creates bottlenecks, worsened by geopolitical tensions and concentrated sourcing.
- Existing studies lack comprehensive, nation-specific analyses of material supply chain constraints and are often prediction-based, lacking real-world applicability.

Research Questions:

- ☑ **What critical materials and volumes are needed for wind and solar PV?**
- ☑ **Can the US meet its net-zero milestones by optimizing available materials? What are the bottlenecks?**

Contributions

- **Material Identification and Quantification:** Identifies critical materials and maps their supply chain flows for wind and solar PV.
- **Optimization Framework:** Creates a decision-making multi-annual optimization model to simulate clean energy planning under material and trade constraints through 2050.
- **Supply Chain Bottlenecks:** Identifies and quantifies supply chain constraints affecting clean energy goals.

Identification and Quantification of Critical Materials

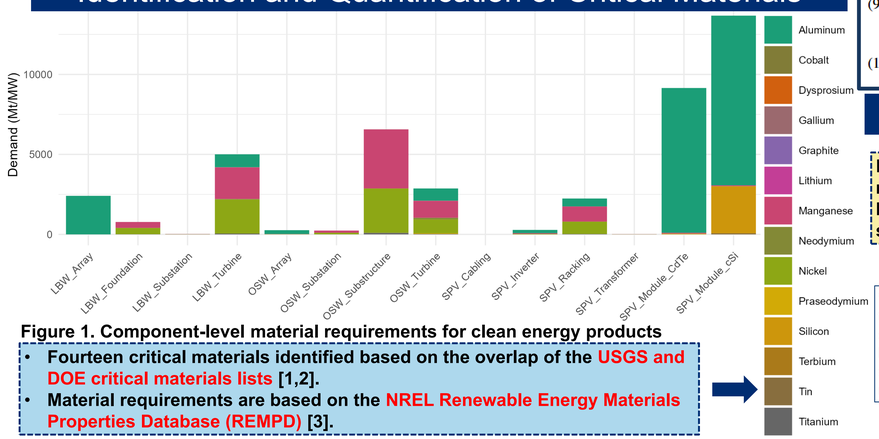


Figure 1. Component-level material requirements for clean energy products

- **Fourteen critical materials identified based on the overlap of the USGS and DOE critical materials lists [1,2].**
- **Material requirements are based on the NREL Renewable Energy Materials Properties Database (REMPD) [3].**

Optimization Model Framework

Optimization Framework: Simulates clean energy deployment decisions over a multi-annual horizon with the following key features.

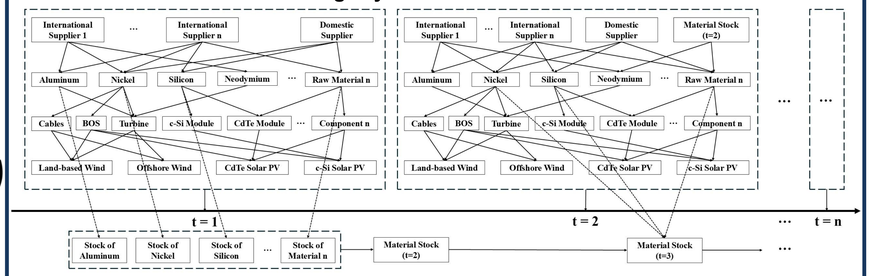


Figure 3. Schematic of multi-annual for material supply and product assembly process

$$(1)\quad \min \sum_{t\in\mathcal{T}} \frac{1}{(1+D)^{t-1}} \left(\sum_{m\in\mathcal{M}} C^{\mathrm{rm}}_{m,t} \cdot \left(\sum_{o\in\mathcal{I}} p_{o,m,t} + \sum_{o\in\mathcal{D}} \frac{q_{o,m,t}}{R^{\mathrm{c}}_{o,m,t}} \right) + \sum_{p\in\mathcal{P}} C^{\mathrm{p}}_{p,t} \cdot x_{p,t} \right)$$

Objective (1): Minimize total discounted costs for material procurement, production, and deployment.

$$(2)\quad y_{m,t} \geq \sum_{c\in\mathcal{C}} z_{c,t} \cdot D^{\mathrm{c}}_{m,c}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(3)\quad z_{c,t} \geq \sum_{p\in\mathcal{P}} x_{p,t} \cdot D^{\mathrm{p}}_{c,p}, \quad \forall c \in \mathcal{C},\ t \in \mathcal{T}$$

Supply Chain Process (2-3): Ensure sufficient materials for components; sufficient components for products.

$$(4)\quad p_{o,m,t} \leq Y_{o,m,t}, \quad \forall o \in \mathcal{I},\ m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(5)\quad q_{o,m,t} \leq R^{\mathrm{c}}_{o,m,t} \cdot Y_{o,m,t}, \quad \forall o \in \mathcal{D},\ m \in \mathcal{M},\ t \in \mathcal{T}$$

Material Acquisition (4-6): Ensure material procurement and domestic production stay within supplier limits; total usage remains within combined supply and stock levels.

$$(6)\quad y_{m,t} \leq R^{\mathrm{d}}_{m,t} \left(\sum_{o\in\mathcal{I}} R^{\mathrm{c}}_{o,m,t} \cdot p_{o,m,t} + \sum_{o\in\mathcal{D}} q_{o,m,t} \right) + s_{m,t}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(7)\quad s_{m,t} = s_{m,t-1} + \left(\sum_{o\in\mathcal{I}} R^{\mathrm{c}}_{o,m,t} \cdot p_{o,m,t-1} + \sum_{o\in\mathcal{D}} q_{o,m,t-1} - y_{m,t-1} \right), \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}$$

Stock Management (7): Ensure stock levels are updated annually based on the balance of procurement, production, and consumption.

$$(8)\quad \underline{X}^{\mathcal{K}}_{t} \leq \sum_{p\in\mathcal{K}} x_{p,t} \leq \overline{X}^{\mathcal{K}}_{t}, \quad \forall \mathcal{K} \in \mathcal{K},\ t \in \mathcal{T}$$

$$(9)\quad \sum_{p\in\mathcal{K}} \left(X_{p,0} + \sum_{\tau=1}^{t} x_{p,\tau} \right) \geq G^{\mathcal{K}}_{t}, \quad \forall \mathcal{K} \in \mathcal{K},\ t \in \mathcal{T}$$

Deploy. Limits & Goals (8-9): Ensure deployment stays within soft limits and meets cumulative goals over the planning horizon.

$$(10)\quad p_{o,m,t},\ q_{o,m,t},\ s_{m,t},\ x_{p,t},\ y_{m,t},\ z_{c,t} \geq 0, \quad \forall o \in \mathcal{I} \cup \mathcal{D},\ m \in \mathcal{M},\ p \in \mathcal{P},\ c \in \mathcal{C},\ t \in \mathcal{T}$$

Non-negativity (10)

Projected Supply of Critical Materials

Project future material availability based on the last supercycle*.

***Supercycle**: Long-term price and demand trends driven by global technological advancements and resource consumption.

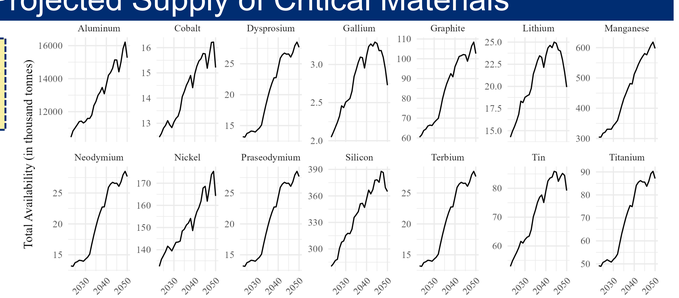


Figure 2. Projected future availability of critical materials from the supercycle

Case Study: Meeting US 2050 Net-Zero Goals

Model Inputs:

1. **Material requirement for identified clean energy products (Fig. 1)** ✚
2. **Projected future critical material availability (Fig. 2)** ✚
3. **Regulated clean energy goals with soft limits (Fig. 3)**
   - DOE targets
   - NREL studies
   - Net-zero roadmaps

Optimization

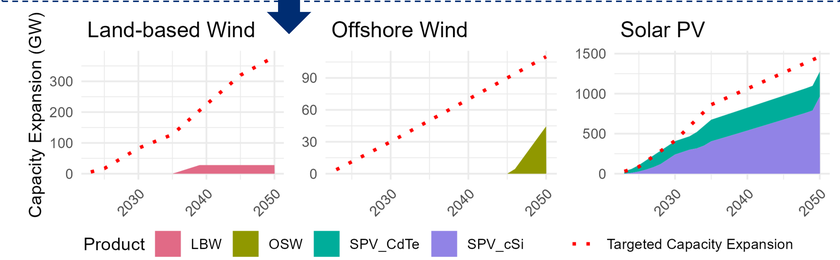
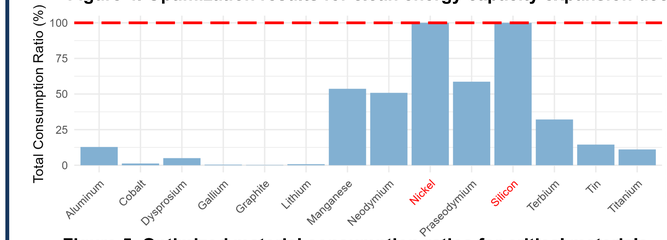


Figure 4. Optimization results for clean energy capacity expansion decisions until 2050



Nickel and silicon reach their availability limits, constraining further deployment of clean energy technologies.

Figure 5. Optimized material consumption ratios for critical materials

Conclusion

- Critical materials like nickel and silicon face severe shortages, limiting clean energy deployment.
- Current material accessibility is insufficient to meet 2050 clean energy goals, with about 1/3 of goals unmet under conservative estimates.
- The US heavily relies on imports, exposing supply chains to geopolitical risks.
- Solutions include boosting domestic production, recycling, material-efficient technologies, and strategic stockpiles.

Ongoing Extensions of the Framework

- **Understanding Supply Chain Constraints for the U.S. Clean Energy Transition** Expanded to include lithium-ion batteries and a broader set of wind and solar technologies. Geopolitical risk is incorporated into the case study. ***Currently under review.***
- **Integrating Upstream Supply Chains into Generation Expansion Planning** Links supply chain decision-making with traditional generation expansion models. ***Submission in progress.***

References:

[1] U.S. Geological Survey, 2024, Mineral commodity summaries 2024: U.S. Geological Survey, 212 p., https://doi.org/10.3133/mcs2024.
[2] Bauer, D. J., Khazdozian, H., Mehta, J., Nguyen, R. T., Severson, M. H., Vaagensmith, B. C., ... & Quaresima, J. (2023). 2023 Critical Materials Strategy (No. INL/RPT-23-72323-Rev. 001). Idaho National Laboratory (INL), Idaho Falls, ID (United States).
[3] NREL (National Renewable Energy Laboratory). 2023. "Renewable Energy Materials Properties Database." Version 1.1. Washington, D.C.: U.S. Department of Energy. https://apps.openei.org/REMPD/.